General Residential Zone (GRZ) Land Development Capacity

SQL Structure

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1. Overview

This SQL script implements a **land development capacity (LDCAP) model for GRZ (General Residential Zone)** properties.

Purpose

- Identify **vacant, infill, and fully developed properties**
- Estimate **development potential (PHU yield / lots)**
- Consider:
  - Planning overlays
  - Hazards
  - Building footprints
  - Access (road frontage)
  - Geometric constraints

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2. Master Orchestration Procedure

property.create_mv_property_grz_ldcap()

This is the **entry point**.

What it does:

1. Drops all dependent materialized views
2. Rebuilds them in correct dependency order

Execution order:

```
building_quadrant_stats
→ building_quadrant_summary
→ viability_overlay_analysis
```

```
4  → infill_viability_testing
5  → vacant_viability_testing
6  → fully_developed_testing
7  → infill_viable
8  → vacant_viable
9  → fully_developed
10  → summary
11  → summary_suburbs
```

Key insight

This ensures:

- All downstream tables always use **fresh upstream outputs**
- Dependencies are correctly sequenced

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3. Core Spatial Analytics

3.1 Building Quadrant Analysis

building_quadrant_stats

Splits each property into **4 quadrants (NE, NW, SE, SW)**.

Process:

1. Select GRZ properties
2. Compute:
    - centroid
    - bounding box
3. Create vertical and horizontal split lines
4. Split polygon into 4 quadrants
5. Assign quadrant IDs (1–4)

Adds:

- **Frontage detection**
    - Based on intersection with road frontage
- **Building overlap**
    - Computes how buildings intersect each quadrant
- **Buildable quadrant**
    - TRUE if:

- no buildings
- frontage exists

Output includes:

- quadrant_geom
- frontage_length
- quadrant_is_front_facing
- overlap_area, building_area
- is_buildable_quadrant

3.2 Quadrant Summary

building_quadrant_summary

Aggregates per property:

- Total buildings
- Total building area
- Building count per quadrant (NE/NW/SE/SW)
- Number of buildable quadrants
- Front-facing quadrant
- Total frontage

Adds:

- **property geometry**

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4. Viability Overlay Analysis (Core Engine)

property_grz_ldcap_viability_overlay_analysis

This is the **central analytical dataset**.

What it calculates:

1. Property metrics

- Area
- Road frontage
- Vacancy flag

2. Overlay constraints (via helper function)

Each returns:

- area
- percentage overlap

Includes:

- Planning layers:
  - designations
  - open space zone
- Infrastructure:
  - facilities
  - powerlines
- Hazards:
  - flood risk A/B
  - slope ≥ 20°
  - liquefaction
  - depressions

3. Combined constraint geometry

```
overlay_geom = UNION of all constraints
```

4. Developable land

```
developable_geom = property - overlay_geom
```

Also calculates:

- developable_area
- developable_percentage

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5. Reusable Function

general.calc_overlap_stats_record(...)

Purpose

Reusable function to:

- intersect property with any dataset
- return:

- overlap geometry
- overlap area
- overlap %

Key features:

- dynamic SQL
- geometry validation
- safe handling of empty cases
- supports filtering (where_clause)

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6. Viability Pipelines

6.1 Vacant Pipeline

Step 1: vacant_viability_testing

Filters properties by excluding:

- facilities / open space
- small lots (<400 or <800 sqm overlay logic)
- overlapping properties
- known bad geometries
- recent building consents

Then:

- removes overlay constraints
- runs **maximum inscribed circle (MIC)**

Criteria:

- radius ≥ 7m
- sufficient area after constraints

Step 2: vacant_viable

Calculates:

Potential lots:

```
floor(developable_area / 400)
```

```
2  or /800 if overlay applies
```

PHU yield logic:

- 1 lot → 1
- 2–6 → same
- ≥7:
    - frontage ≥10m → 70% yield
    - else → capped at 6

Access constraint:

- frontage ≥ 3.6m minimum

6.2 Infill Pipeline

Step 1: infill_viability_testing

More complex than vacant.

Additional filtering:

- building coverage ≥ 25%
- multiple buildings
- setbacks
- access width (≥3.6m)
- centroid conflicts

Geometry processing:

1. remove overlays
2. remove buildings (+ buffer)
3. check remaining usable land

Advanced checks:

- setback clearance
- access side count
- building quadrant logic

Platform test:

- MIC radius ≥ 7m
- minimum usable space

Step 2: infill_viable

Same structure as vacant but:

PHU yield differs:

- 1 lot → 0 (no gain)
- 2–6 → lots - 1
- ≥7:
    - frontage ≥10m → reduced (70%)
    - else → capped at 6

6.3 Fully Developed

fully_developed

Defined as:

- GRZ properties not:
    - vacant
    - infill viable
    - excluded overlays

Output:

- 1 lot only
- 0 PHU yield

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7. Summary Outputs

7.1 Global Summary

property_grz_ldcap_summary

Aggregates:

- number of properties
- total lots
- total PHU
- total area

By:

- development type:
  - Vacant / Infill
- scale:
  - Single
  - Small (2–6)
  - Large (7+)

7.2 Suburb Summary

property_grz_ldcap_summary_suburbs

Same as above but grouped by:

- suburb (SA2)

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8. Exclusion Threshold Framework

Table:

property_grz_ldcap_exclusion_thresholds

Purpose:

Data-driven rule system for excluding properties.

Supports:

- operators: > >= < <= = BETWEEN
- linked to:
  - overlay percentages

Example rules:

- slope > 30%
- flood ≥ 30%
- liquefaction ≥ 30%

Used in:

- vacant_viable
- infill_viable

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9. Building Ranking Function

property.property_buildings_ranking(prop_no)

Ranks buildings by likelihood of being the **primary dwelling**.

Criteria:

- area (largest better)
- distance to frontage (closer better)
- alignment with frontage
- back-of-lot penalty
- heuristic classification

Output:

- ranked buildings
- or fallback point if none exist

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10. Key Design Patterns

10.1 Materialized View Pipeline

- All outputs cached
- Full refresh model

10.2 Progressive Filtering

Each stage:

```
raw → testing → viable → summary
```

10.3 Geometry-first logic

Everything derived from:

- spatial intersection
- buffering
- difference
- MIC

10.4 Strong performance design

- Extensive GIST indexes

- lateral joins to avoid recomputation
- shared overlay function

10.5 Config-driven rules

- thresholds table enables tuning without SQL changes

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11. Data Outputs (Core Tables)

Table	Purpose
..._overlay_analysis	Core constraint + developable land
..._vacant_viable	Final vacant sites
..._infill_viable	Final infill sites
..._fully_developed	Non-developable
..._summary	Aggregated totals
..._summary_suburbs	Suburb rollup
..._building_quadrant_stats	Fine-grained building layout
..._building_quadrant_summary	Property-level building stats

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12. End-to-End Conceptual Flow

```
GRZ properties
   ↓
Overlay analysis (constraints + developable land)
   ↓
Branch pipelines:
   ├── Vacant
   ├── Infill
   └── Fully developed
   ↓
Viability filtering
   ↓
Lot estimation + PHU yield
   ↓
Aggregation (global + suburb)
```

Final Takeaway

This script is a **full spatial decision engine** for residential capacity that:

- Combines **planning + hazard + geometry + access**
- Uses **robust spatial modelling (PostGIS-heavy)**
- Applies **clear staged filtering**
- Produces **policy-ready outputs (PHU, development categories)**